

The empirical recurrence for OEIS A267790: recovering a conjecture that is not written down

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Abstract

OEIS A267790 records “Empirical recurrence of order 93”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 589$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 93, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 93 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A267790 is “Number of $4 \times n$ 0..1 arrays with every repeated value in every row greater than or equal to, and in every column greater than, the previous repeated value.”. It has offset 1 and begins

9, 81, 729, 5081, 34173, 211555, 1262760, 7263481, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Jan 20 2016 12:33:12, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 589$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 589$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1178$ exact terms of the model returns order 93 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	7	c_{25}	−29344008349075156	c_{49}	329425916682015277681	c_{73}	−10194991852861
c_2	146	c_{26}	−18277944741772011	c_{50}	180652488123684295482	c_{74}	114536503675562
c_3	−924	c_{27}	132419862238248177	c_{51}	−269387107207855799884	c_{75}	23379882328505
c_4	−10903	c_{28}	179378752758150447	c_{52}	−196892994917529108855	c_{76}	−19589813512821
c_5	56817	c_{29}	−435855189906748073	c_{53}	188062757032999373569	c_{77}	−3352861810889
c_6	544979	c_{30}	−990864030288492155	c_{54}	173665376535217541705	c_{78}	28363830748151
c_7	−2108799	c_{31}	943383768121025491	c_{55}	−112227589008523308213	c_{79}	2516299575288
c_8	−19942537	c_{32}	3882098172348042152	c_{56}	−128264047102178767998	c_{80}	−3429676422380
c_9	50560415	c_{33}	−681325847197356290	c_{57}	57164577400701766526	c_{81}	118011065952
c_{10}	553364590	c_{34}	−11509738813630938430	c_{58}	80735845403434798866	c_{82}	3399174333792
c_{11}	−741204644	c_{35}	−4134478547517963382	c_{59}	−24732377337035497322	c_{83}	−64600007808
c_{12}	−11812169286	c_{36}	26469761422211512097	c_{60}	−43752711423849505039	c_{84}	−268921137600
c_{13}	3651204268	c_{37}	21683131137015469955	c_{61}	8999567560413197299	c_{85}	9965850624
c_{14}	194832320759	c_{38}	−47397215653916159200	c_{62}	20536945053992935060	c_{86}	16327676032
c_{15}	115457988771	c_{39}	−62571985758658690474	c_{63}	−2703022201943272362	c_{87}	−929074304
c_{16}	−2474103390819	c_{40}	64646561557890071759	c_{64}	−8377653489661972711	c_{88}	−714186112
c_{17}	−3566275657511	c_{41}	132024013571983973423	c_{65}	644994341682713645	c_{89}	55635456
c_{18}	23818896644863	c_{42}	−61420192046843542585	c_{66}	2974256924421686355	c_{90}	20027904
c_{19}	56365350982953	c_{43}	−220216698697265644055	c_{67}	−111015798559622211	c_{91}	−1992704
c_{20}	−166854588073303	c_{44}	23950462815731678762	c_{68}	−918769698647895624	c_{92}	−270336
c_{21}	−611369891464329	c_{45}	300669638903518349222	c_{69}	8847935378160904	c_{93}	32768
c_{22}	748544390034390	c_{46}	45216533928671764832	c_{70}	246514350084304134		
c_{23}	4882954026903148	c_{47}	−342533981756723774148	c_{71}	2025312020960226		
c_{24}	−808411093653898	c_{48}	−123566042637634254441	c_{72}	−57257245026017130		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 93, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $93 + 4$ terms removed returns order 93 as well, so $\deg q_{\min} = 93 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $93 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 93 that this sequence satisfies — and that family turns out to have exactly one member.

References

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